

Graph Neural Network:

For Node or graph classification

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Objectives



Reproducibility

Reproduce the original GCN, GraphSAGE, GAT, and GIN models.



Evaluation

Evaluate the models on both original and new graph datasets with accuracy, micro average f1 et macro average f1



Harmonisation

Harmonize selected datasets to improve data quality

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Method overview

Node Classification (GNN) & Evaluation & Harmonisation

GNNs

Definition

Graph Neural Networks (GNNs) are deep learning models designed to learn from graph-structured data by aggregating information from neighboring nodes.

Types of GNNs

- Graph Convolutional Network
- Graph Attention Network
- Graph Isomorphism Network
- GraphSage ...

Idea

Relies on the message-passing paradigm

- Each node passes messages to its neighbors

- AGGREGATE $m_{N(u)}^{(k)} = \text{AGGREGATE}^{(k)}(\{h_v^{(k)} \mid v \in N(u)\})$

- UPDATE $h_u^{(k+1)} = \text{UPDATE}^{(k)}(h_u^{(k)}, m_{N(u)}^{(k)})$

Formula of Nodes representation

$$H^{(k)} = \sigma \left(A H^{(k-1)} W_{\text{neigh}}^{(k)} + H^{(k-1)} W_{\text{self}}^{(k)} \right)$$

GCN

Definition

GCN extends GNNs by performing convolution operations on graph structures.

It aggregates information from neighboring nodes using a normalized adjacency matrix.

Each node representation is updated by combining its own features with neighborhood information.

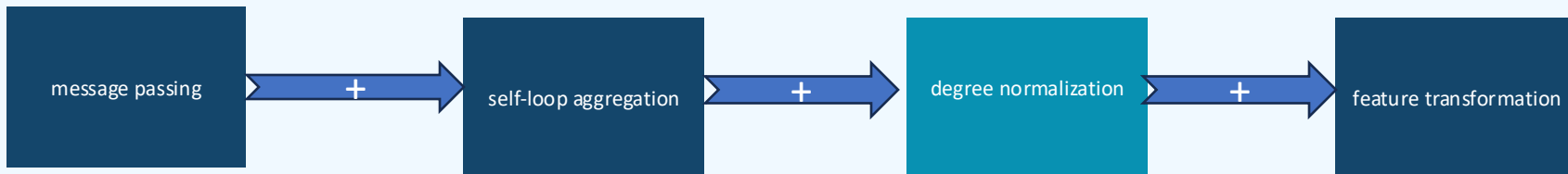
Formula of nodes representation

$$H^{(k)} = \sigma \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(k-1)} W^{(k)} \right)$$

where:

- $\tilde{A} = A + I$ (graph with self-loops),
- $\tilde{D}$ is the degree matrix of $\tilde{A}$,
- $W^{(k)}$ is a learnable weight matrix,
- σ is a nonlinear activation function.

Modeling



GraphSAGE

Definition

GraphSAGE (Graph Sample and Aggregate) is an inductive Graph Neural Network proposed by Hamilton et al. that learns a function to generate node embeddings by aggregating information from local neighborhoods, instead of learning a unique embedding for each node

Embedding generation algorithm

Input : Graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$; input features $\{\mathbf{x}_v, \forall v \in \mathcal{V}\}$; depth K ; weight matrices $\mathbf{W}^k, \forall k \in \{1, \dots, K\}$; non-linearity σ ; differentiable aggregator functions $\text{AGGREGATE}_k, \forall k \in \{1, \dots, K\}$; neighborhood function $\mathcal{N} : v \rightarrow 2^{\mathcal{V}}$

Output : Vector representations $\mathbf{z}_v$ for all $v \in \mathcal{V}$

```
1  $\mathbf{h}_v^0 \leftarrow \mathbf{x}_v, \forall v \in \mathcal{V}$ ;
2 for  $k = 1 \dots K$  do
3   for  $v \in \mathcal{V}$  do
4      $\mathbf{h}_{\mathcal{N}(v)}^k \leftarrow \text{AGGREGATE}_k(\{\mathbf{h}_u^{k-1}, \forall u \in \mathcal{N}(v)\})$ ;
5      $\mathbf{h}_v^k \leftarrow \sigma(\mathbf{W}^k \cdot \text{CONCAT}(\mathbf{h}_v^{k-1}, \mathbf{h}_{\mathcal{N}(v)}^k))$ 
6   end
7    $\mathbf{h}_v^k \leftarrow \mathbf{h}_v^k / \|\mathbf{h}_v^k\|_2, \forall v \in \mathcal{V}$ 
8 end
9  $\mathbf{z}_v \leftarrow \mathbf{h}_v^K, \forall v \in \mathcal{V}$ 
```

Aggregator variants

- Mean Aggregator

$$h_{N(v)}^{(k)} = \frac{1}{|N(v)|} \sum_{u \in N(v)} h_u^{(k-1)}$$

- GCN Aggregator

$$h_v^{(k)} = \sigma(W^{(k)} \cdot \text{MEAN}(\{h_v^{(k-1)}\} \cup \{h_u^{(k-1)}, u \in N(v)\}))$$

- LSTM Aggregator

$$h_{N(v)}^{(k)} = \text{LSTM}(\{h_u^{(k-1)}, u \in N(v)\})$$

- Pooling Aggregator

$$\text{AGGREGATE}_{pool}^{(k)} = \max(\{\sigma(W_{pool} h_u^{(k)} + b), \forall u \in N(v)\})$$

GraphSAGE vs. GCN: Key Differences

- Inductive representation learning
- scalable to large graphs through neighborhood sampling
- supports unseen nodes and unseen graphs
- flexible aggregation architectures

GAT

Definition

GAT extends GCN by introducing a learnable attention mechanism.

Each neighbor receives an importance weight during message aggregation.

Unlike GCN, all neighbors are not treated equally.

Attention coefficient formula

$$\alpha_{u,v} = \frac{\exp(a^\top [Wh_u || Wh_v])}{\sum_{v' \in N(u)} \exp(a^\top [Wh_u || Wh_{v'}])}$$

Node representation computation

$$h_u^{(k)} = \sigma \left(\sum_{v \in N(u)} \alpha_{u,v} Wh_v^{(k-1)} \right)$$

where $N(u)$ includes node u itself

GAT vs. GCN: Key Differences

- GCN: fixed normalization weights based on graph structure,
- GAT: adaptive weights learned through attention

GIN

Definition

GIN aims to maximize the expressive power of Graph Neural Networks. It uses sum aggregation to better distinguish different graph structures. Unlike GCN/GAT, GIN combines neighborhood aggregation with an MLP.

Graph representation computation

$$h_G = \text{CONCAT}(\text{READOUT}(\{h_v^{(k)} | v \in G\}))$$

READOUT: aggregates the node embeddings of a graph using summation.

CONCAT: concatenates the graph representations from all GIN layers to capture both local and global structural information.

Node representation computation

$$h_u^{(k)} = \text{MLP}^{(k)} \left((1 + \epsilon^{(k)})h_u^{(k-1)} + \sum_{v \in N(u)} h_v^{(k-1)} \right)$$

ϵ controls the relative importance of the node's own features and its neighbors' features.

GIN VS others GNNs : Key Differences

- **Injective aggregation:** Sum aggregation enables GIN to match the discriminative power of the Weisfeiler-Lehman test.
- **MLP-based update:** An MLP improves the expressiveness of node representations after aggregation.
- **Multi-layer readout:** Combines representations from all layers to capture local and global graph structures.

Datasets

Citation Networks (Node Classification — GCN & GAT)

Cora (2708 nodes, 5278 edges, 1433 features, 7 classes)

CiteSeer (3327 nodes, 4552 edges, 3703 features, 6 classes)

PubMed (19717 nodes, 44324 edges, 500 features, 3 classes)

Graph Classification Benchmarks (GIN)

IMDB-BINARY (1000 graphs, 2 classes, 19.8 avg. nodes)

IMDB-MULTI (1500 graphs, 3 classes, 13.0 avg. nodes)

REDDIT-BINARY (2000 graphs, 2 classes, 429.6 avg. nodes)

REDDIT-MULTI-5K (5000 graphs, 5 classes, 508.5 avg. nodes)

COLLAB (5000 graphs, 3 classes, 74.5 avg. nodes)

MUTAG (188 graphs, 2 classes, 17.9 avg. nodes)

PROTEINS (1113 graphs, 2 classes, 39.1 avg. nodes)

PTC (344 graphs, 2 classes, 25.5 avg. nodes)

NCI1 (4110 graphs, 2 classes, 29.8 avg. nodes)

Additional Real-world and Heterophilic Datasets (GCN & GAT)

Cornell (183 nodes, 280 edges, 5 classes)

Texas (183 nodes, 295 edges, 5 classes)

Wisconsin (251 nodes, 466 edges, 5 classes)

Chameleon (2277 nodes, 31421 edges, 5 classes)

Squirrel (5201 nodes, 198493 edges, 5 classes)

Actor (7600 nodes, 26752 edges, 5 classes)

Roman-Empire (22662 nodes, 32927 edges, 18 classes)

Amazon-Ratings (24492 nodes, 93050 edges, 5 classes)

Minesweeper (10000 nodes, 39402 edges, 2 classes)

Tolokers (11758 nodes, 519000 edges, 2 classes)

Questions (48921 nodes, 153540 edges, 2 classes)

Metrics

Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

f1-micro average

$$\text{Micro-F1} = \frac{2 \times \sum TP}{2 \times \sum TP + \sum FP + \sum FN}$$

f1-macro average

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^C F1_i$$

where C is the number of classes and F1_i denotes the F1-score of class i.

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Evaluation & Results

23 datasets · node classification models (GAT/GCN/GIN/GraphSAGE) · 3 metric

Results — Accuracy&f1-micro&f1-macro ↑-GCN

Dataset	GCN1 (%)	GCN2 (%)	F1-micro	F1-macro
	Accuracy	Accuracy		
Cora	81.50	81.60	81.60	80.63
Citeseer	70.30	71.70	71.70	68.96
PubMed	79.00	78.60	78.60	78.21
Cornell	--	78.20	78.20	46.14
Texas	--	70.83	70.83	39.35
Wisconsin	--	66.36	66.36	54.42
Chameleon	--	48.94 (46.84)	48.94 (46.84)	47.10 (46.26)
Squirrel	--	32.30 (31.96)	32.30 (31.96)	31.66 (31.14)
Actor	--	36.81	36.81	33.05
Roman-Empire	--	25.59	25.59	15.29
Amazon-Ratings	--	38.17	38.17	17.80
Minesweeper	--	80.13	80.13	45.91
Tolokers	--	78.20	78.20	46.14
Questions	--	97.03	97.03	49.25

- GCN1: Results reported by Kipf and Welling (2017)). And GCN2: results obtained from our implementation
- Values in parentheses correspond to results after removing near-duplicate nodes for Chameleon and Squirrel.

Results — f1-micro&f1-macro ↑-GraphSAGE

Dataset	Agg	GraphSAGE1 (%)		GraphSAGE2 (%)	
		f1-micro	f1-micro	f1-micro	f1-macro
PPI	GCN	50.00	51.13		30.33
	Mean	59.80	57.62		41.19
	LSTM	61.20	60.29		45.40
	Pool	60.00	59.11		43.35
Cora	GCN	--	81.40		80.12
	Mean	--	83.20		82.54
	LSTM	--	80.20		78.24
	Pool	--	81.20		80.85
Citeseer	GCN	--	73.00		67.25
	Mean	--	70.00		64.74
	LSTM	--	68.80		63.77
	Pool	--	69.00		63.85
PubMed	GCN	--	86.40		85.58
	Mean	--	90.00		89.78
	LSTM	--	89.60		89.14
	Pool	--	88.80		88.37
Cornell	GCN	--	67.80		49.87
	Mean	--	71.19		52.16
	LSTM	--	77.97		57.39
	Pool	--	64.41		48.33
Texas	GCN	--	64.41		35.55
	Mean	--	79.66		48.23
	LSTM	--	81.36		56.97
	Pool	--	66.10		38.85

GraphSAGE1: results reported by Hamilton et al. (2017). And GraphSAGE2: results obtained from our implementation

Results — f1-micro&f1-macro ↑-GraphSAGE

Dataset	Agg	GraphSAGE1 (%) f1-micro	GraphSAGE2 (%) f1-micro	f1-macro
Wisconsin	GCN	--	61.25	35.53
	Mean	--	75.00	44.05
	LSTM	--	77.50	48.37
	Pool	--	55.00	28.28
Actor	GCN	--	27.84	23.80
	Mean	--	33.43	30.70
	LSTM	--	33.35	30.49
	Pool	--	34.05	28.43
Roman-Empire	GCN	--	44.84	35.23
	Mean	--	71.24	62.88
	LSTM	--	71.37	63.60
	Pool	--	73.65	67.20
Amazon-Ratings	GCN	--	43.43	34.44
	Mean	--	45.47	36.94
	LSTM	--	43.21	35.54
	Pool	--	42.58	27.33
Minesweeper	GCN	--	80.00	44.44
	Mean	--	81.48	65.86
	LSTM	--	81.60	64.00
	Pool	--	82.04	65.79
Tolokers	GCN	--	77.82	46.20
	Mean	--	78.53	57.31
	LSTM	--	78.22	52.70
	Pool	--	78.26	44.36

GraphSAGE1: results reported by Hamilton et al. (2017). And GraphSAGE2: results obtained from our implementation

Results — f1-micro&f1-macro ↑-GraphSAGE

Dataset	Agg	GraphSAGE1 (%)		GraphSAGE2 (%)
		f1-micro	f1-micro	f1-macro
Questions	GCN	--	97.05	53.32
	Mean	--	96.25	58.42
	LSTM	--	96.35	58.67
	Pool	--	96.84	56.87
Chameleon	GCN	--	40.60 (47.78)	40.68 (46.14)
	Mean	--	52.81 (51.78)	52.48 (50.00)
	LSTM	--	49.38 (55.92)	47.99 (55.47)
	Pool	--	42.94 (46.89)	41.83 (44.84)
Squirrel	GCN	--	32.81 (34.84)	30.23 (33.58)
	Mean	--	37.20 (37.61)	36.77 (37.45)
	LSTM	--	34.38 (35.74)	33.77 (35.33)
	Pool	--	35.64 (38.76)	33.87 (36.40)

- GraphSAGE1: results reported by Hamilton et al. (2017). And GraphSAGE2: results obtained from our implementation
- Values in parentheses for Chameleon and Squirrel correspond to the results obtained after removing near-duplicate nodes

Results — Accuracy & f1-micro & f1-macro ↑ -GAT

Dataset	GAT1 (%)	GAT2 (%)		
	acc	acc	f1-micro	f1-macro
Cora	83.00	83.40	83.40	82.35
Citeseer	72.50	73.20	73.20	67.38
PubMed	79.00	--	--	--
Cornell	--	45.40	45.40	15.37
Texas	--	55.36	55.36	14.25
Wisconsin	--	52.99	52.99	20.61
Chameleon	--	57.40 (57.20)	57.40 (57.20)	55.53 (55.45)
Squirrel	--	36.03 (36.95)	36.03 (36.95)	31.05 (35.56)
Actor	--	26.96	26.96	13.59

- GAT1: results reported by Veličković et al. (2018). And GAT2: results obtained from our implementation
- Values in parentheses correspond to results after removing near-duplicate nodes for Chameleon and Squirrel.

Results — Accuracy&f1-micro&f1-macro ↑-GIN

Dataset	GIN1 (%)	GIN2 (%)		
	Accuracy	Accuracy	F1-micro	F1-macro
MUTAG	89.4 ± 5.6	93.04 ± 4.90	93.04 ± 4.90	92.39 ± 5.17
PTC	64.6 ± 7.0	72.08 ± 4.93	72.08 ± 4.93	69.69 ± 6.83
NCI1	82.7 ± 1.7	78.03 ± 1.48	78.03 ± 1.48	78.01 ± 1.48
PROTEINS	76.2 ± 2.8	79.51 ± 3.31	79.51 ± 3.31	78.49 ± 2.94
IMDBBINARY	75.1 ± 5.1	74.80 ± 2.44	74.80 ± 2.44	74.59 ± 2.29
IMDBMULTI	52.3 ± 2.8	50.73 ± 1.59	50.73 ± 1.59	48.57 ± 2.84
REDDITBINARY	92.4 ± 2.5	78.20 ± 1.95	78.20 ± 1.95	78.15 ± 1.96
COLLAB	80.2 ± 1.9	-- ± --	-- ± --	-- ± --
REDDIT-MULTI5K	57.5 ± 1.5	-- ± --	-- ± --	-- ± --

- GIN1: results reported by Xu et al. (2019) using GIN- ϵ with $\epsilon = 0$
- GIN2: results obtained from our implementation using 10-fold cross-validation.
- Reported values correspond to the mean performance $\pm$ standard deviation over the folds

Comparative Study of Three GNNs

Criterion	GCN	GraphSAGE	GAT	GIN
Original learning setting	Transductive	Inductive	Transductive	Graph and Node classification
Aggregation	Normalized mean	Mean / Max / LSTM	Attention	Injective sum
Attention mechanism	X	X	✓	X
MLP after aggregation	X	X	X	✓
Scalability	Medium	High	Low	Medium
Training speed	Fast	Very fast	Slow	Medium
Expressive power	Medium	Medium	High	Highest (1-WL)

Thank you for your attention

Questions ?